(2) A local minorization condition on the sub-level sets of the Lyapunov function.

Such drift-minorization approaches are well-established techniques to localize the minorization conditions on the compact sub-level sets of the Lyapunov function. In this context, Markov operators act on weighted Banach spaces and operator norms coincide with the contraction coefficients of the Markov integral operator. For a more detailed discussion on this contraction coefficients we refer to [6, 39, 41, 44].

This Lyapunov approach and the operator-theoretic framework presented in this article strongly differ from the approaches discussed earlier and taken in the series of recent articles [22, 26, 45, 54]. Our main contributions can be summarized in the following way:

- *Strict contraction for bounded costs:* For bounded transition potentials, we establish strict contraction inequalities for a broad class of ϕ-divergences, including $\mathbb{L}_p$ norms, Havrda–Charvat and α-divergences, Jensen–Shannon divergence, squared Hellinger and Kakutani–Hellinger distances, as well as Jeffreys and Rényi divergences (see Theorem 2.2 and Section 4.1).

- *Weighted-norm stability for unbounded models:* For locally bounded costs and marginal potentials with compact sub-level sets, we introduce weighted total variation and Kantorovich–Wasserstein contraction estimates, together with relative entropy bounds for Schrödinger bridges (Sections 4.4, 4.5, and 4.6). The choice of Lyapunov functions and Wasserstein metrics is explicitly adapted to the marginal distributions.

- *General Lyapunov contraction principles:* We formulate practical drift–minorization conditions yielding quantitative exponential convergence rates for a large class of Sinkhorn semigroups. These results apply to models with polynomial growth, heavy-tailed marginals, boundary state spaces, semi-compact domains, and finite mixture models, including kernel-based density estimators used in generative modeling.

To the best of our knowledge, these results provide the first general exponential stability estimates for Sinkhorn semigroups with respect to weighted norms, ϕ-divergences, and Wasserstein-type distances in non-necessarily compact and unbounded settings.

1.4. **Outline of the article.** After introducing notation in Section 1.5, Section 2 recalls the Schrödinger bridge problem and the associated Sinkhorn semigroups, emphasizing their interpretation as time-inhomogeneous nonlinear Markov evolutions (see also [2]).

Section 2.3 presents a concise overview of the main stability and contraction results established in this work. Motivating applications are already discussed in Section 2.4, which highlights the relevance of our approach for non-compact state spaces, unbounded costs, and heavy-tailed marginals.

Sections 3–3.2 develop the operator-theoretic foundations underlying our analysis. We review Lyapunov drift–minorization techniques and weighted Banach space contractions for Markov integral operators (Section 3, with complements in Appendix C), recall the Hilbert projective metric and ϕ-divergences (Section 3.1), and introduce the associated contraction coefficients (Section 3.2).

Building on this framework, Section 4 adapts weighted-norm and Lyapunov contraction techniques to Sinkhorn semigroups. Strong mixing regimes are analyzed in Section 4.1, where we establish strict contraction properties for a broad class of ϕ-divergences, including relative entropy, total variation, $\mathbb{L}_p$ norms, Hellinger-type divergences, Jensen–Shannon, Jeffreys, and Rényi divergences. General drift–minorization conditions ensuring quantitative contraction bounds are developed in Section 4.2. The main exponential stability theorems are presented in Section 4.3.

Section 4.4 introduces a simple and practically verifiable Lyapunov condition yielding exponential convergence on weighted Banach spaces. Extensions to Wasserstein-type semi-metrics are derived in Section 4.5, while Section 4.6 is devoted to relative-entropy decay and the convergence of Sinkhorn bridges.

Section 5 illustrates the scope of the theory through a wide range of examples, including polynomial growth models (Section 5.1), heavy-tailed marginals (Section 5.2), bounded Lipschitz domains (Section 5.3), Beta and Weibull models (Sections 5.4 and 5.5), and multivariate linear Gaussian models (Section 5.8). Section 5.9 extends the analysis to finite mixture models, including non-Gaussian and kernel-based constructions relevant to generative modeling.

Detailed proofs and complementary material are collected in the appendices. Appendix A focuses on Gaussian Sinkhorn algorithms, Appendices B and C provide additional results on Hilbert metrics and weighted-norm contractions, Appendix D reviews Schrödinger potentials, and Appendix E contains further technical arguments.

1.5. **Some basic notation.** We let $\mathcal{B}(\mathbb{X})$ be the algebra of locally bounded measurable functions on a locally compact Polish space $\mathbb{X}$. We denote by $\mathcal{B}_b(\mathbb{X}) \subset \mathcal{B}(\mathbb{X})$ the sub-algebra of bounded measurable functions f on $\mathbb{X}$ endowed with the supremum norm $\|f\| := \sup_{x\in\mathbb{X}} |f(x)|$. For a given uniformly positive function $\varphi \in \mathcal{B}(\mathbb{X})$, we let $\mathcal{B}_\varphi(\mathbb{X}) \subset \mathcal{B}(\mathbb{X})$ be the Banach space of functions $f \in \mathcal{B}(\mathbb{X})$ equipped with the norm $\|f\|_\varphi := \|f/\varphi\|$.

Let $\mathcal{M}(\mathbb{X})$ be the set of locally bounded signed measures on $\mathbb{X}$, and let $\mathcal{M}_b(\mathbb{X}) \subset \mathcal{M}(\mathbb{X})$ the subset of bounded measures. We denote the convex subset of probability measures on $\mathbb{X}$ as $\mathcal{M}_1(\mathbb{X}) \subset \mathcal{M}_b(\mathbb{X})$.

The Lebesgue integral of an integrable function $f \in \mathcal{B}(\mathbb{X})$ w.r.t. $\nu \in \mathcal{M}(\mathbb{X})$ is denoted by

$$\nu(f) := \int_{\mathbb{X}} f(x)\ \nu(dx).$$

For indicator functions $f = 1_A$ of a measurable subset $A \subset \mathbb{X}$ sometimes we write $\nu(A)$ instead of $\nu(1_A)$.

Consider a bounded integral operator $\mathcal{K}(x, dy)$ from $\mathbb{X}$ into $\mathbb{Y}$, a measure $\mu \in \mathcal{M}(\mathbb{X})$ and a function $f \in \mathcal{B}(\mathbb{Y})$. We denote by $\mu\mathcal{K} \in \mathcal{M}(\mathbb{Y})$ and $\mathcal{K}(f) \in \mathcal{B}(\mathbb{X})$ the measure and the function defined by

$$(\mu\mathcal{K})(dy) := \int \mu(dx)\mathcal{K}(x, dy) \quad \text{and} \quad \mathcal{K}(f)(x) := \int \mathcal{K}(x, dy)f(y). \tag{1.6}$$

For indicator functions $f = 1_A$ sometimes we write $\mathcal{K}(x, A)$ instead of $\mathcal{K}(1_A)(x)$. For bounded integral operator $\mathcal{K}_1(x, dy)$ from $\mathbb{X}$ into $\mathbb{Y}$ and $\mathcal{K}_2(y, dx)$ from $\mathbb{Y}$ into $\mathbb{X}$, we also denote by $(\mathcal{K}_1\mathcal{K}_2)$ the bounded integral operator from $\mathbb{X}$ into $\mathbb{X}$ defined by the integral composition

$$(\mathcal{K}_1\mathcal{K}_2)(x_1, dx_2) := \int \mathcal{K}_1(x_1, dy)\mathcal{K}_2(y, dx_2).$$

Given a Markov transition $\mathcal{K}(x, dy)$ from $\mathbb{X}$ into $\mathbb{Y}$ and a measure $\mu \in \mathcal{M}(\mathbb{X})$ we set

$$(\mu \times \mathcal{K})(d(x, y)) := \mu(dx)\,\mathcal{K}(x, dy) \quad \text{and} \quad (\mu \times \mathcal{K})^\flat(d(y, x)) := \mu(dx)\,\mathcal{K}(x, dy).$$

A measure $\nu_1 \in \mathcal{M}(\mathbb{X})$ is said to be absolutely continuous with respect to another measure $\nu_2 \in \mathcal{M}(\mathbb{X})$ and we write $\nu_1 \ll \nu_2$ if $\nu_2(A) = 0$ implies $\nu_1(A) = 0$ for any $A \in \mathbb{X}$. Whenever $\nu_1 \ll \nu_2$, we denote by $d\nu_1/d\nu_2$ the Radon-Nikodym derivative of ν_1 w.r.t. ν_2. We also consider the partial order, denoted $\nu_1 \leqslant \nu_2$, when $\nu_1(A) \leqslant \nu_2(A)$ for any measurable subset $A \subset \mathbb{X}$. The measures are said comparable when we have $c\ \nu_1 \leqslant \nu_2 \leqslant c^{-1}\ \nu_1$ for some scalar constant $c > 0$.

The positive and negative part of $a \in \mathbb{R}$ are defined respectively by $a_+ = \max(a, 0)$ and $a_- = \max(-a, 0)$. Given $a, b \in \mathbb{R}$ we also set $a \vee b = \max(a, b)$ and $a \wedge b = \min(a, b)$.

Kantorovich and Wasserstein criteria. We consider the weighted total variation norm associated with some positive function $\varphi \in \mathcal{B}(\mathbb{X})$ defined as

$$\|\!|\mu_1 - \mu_2|\!\|_{\varphi} := \inf_{\mathcal{P}\in\Pi(\mu_1,\mu_2)} \int (\varphi(x_1) + \varphi(x_2))\ 1_{x_1 \neq x_2}\ \mathcal{P}(d(x_1,x_2)), \tag{1.7}$$

where $\Pi(\mu_1,\mu_2)$ stands for the set of coupling probability measures $\mathcal{P}$ on the product space $(\mathbb{X}\times\mathbb{X})$, with prescribed first and second coordinate marginals (μ_1,μ_2) such that $\mu_i(\varphi) < \infty$. For a given semi-metric $\varrho(x_1,x_2)$ on $\mathbb{X}$ and $p \geqslant 1$ we also consider the Wasserstein semi-metric $\mathbb{W}_{p,\varrho}$ defined by

$$\mathbb{W}_{p,\varrho}(\mu_1,\mu_2) := \inf_{\mathcal{P}\in\Pi(\mu_1,\mu_2)} \left(\int\ \varrho(x_1,x_2)^p\ \mathcal{P}(d(x_1,x_2))\right)^{1/p}. \tag{1.8}$$

When $p=1$ we simplify notation and we write $\mathbb{W}_{\varrho}$ instead of $\mathbb{W}_{1,\varrho}$. When $\varrho_{\|\cdot\|}(x_1,x_2) := \|x_1 - x_2\|$ is associated with some norm $\|\cdot\|$ on some vector spaces $\mathbb{X}$, when there are no possible confusions we write $\mathbb{W}_p$ instead of $\mathbb{W}_{p,\varrho_{\|\cdot\|}}$.

When $\varrho(x_1,x_2)$ is a metric on $\mathbb{X}$ (possibly different from the metric of the Polish space) we recall (see for instance [55, 57, 70, 102, 104]) the Kantorovich-Rubinstein duality

$$\mathbb{W}_{\varrho}(\mu_1,\mu_2) = \sup\{(\mu_1-\mu_2)(f)\ ;\ f\ :\ \mathbb{X}\mapsto\mathbb{R}\quad s.t.\quad |f(x_1)-f(x_2)| \leqslant \varrho(x_1,x_2)\}.$$

Note that

$$\varrho_{\varphi}(x_1,x_2) = (\varphi(x_1)+\varphi(x_2))\ 1_{x_1\neq x_2} \quad\Longrightarrow\quad \mathbb{W}_{\varrho_{\varphi}}(\mu_1,\mu_2) = \|\!|\mu_1-\mu_2|\!\|_{\varphi}.$$

More generally, for any given semi-metric $\varrho(x_1,x_2)$ on $\mathbb{X}$ such that

$$\varphi(x_1)+\varphi(x_2) \geqslant \varrho(x_1,x_2) \quad\text{and}\quad \varphi(x) \geqslant c > 0 \tag{1.9}$$

we have the estimates

$$\mathbb{W}_{\varrho}(\mu_1,\mu_2) \vee (c\ \|\mu_1-\mu_2\|_{tv}) \leqslant \|\!|\mu_1-\mu_2|\!\|_{\varphi}, \quad \text{with the total variation norm } \|\cdot\|_{tv}. \tag{1.10}$$

For instance on a given normed vector space $(\mathbb{X},\|\cdot\|)$, for any $b,c>0$ and $p,q\geqslant 1$, if

$$\varphi(x) := c\ \exp(b\|x\|^p)$$

then

$$\varphi(x_1)+\varphi(x_2) \geqslant \varrho_{exp}(x_1,x_2) := 2c\ \left(\exp\left(\frac{b}{2^{1+(p-1)_+}}\ \|x_1-x_2\|^p\right)-1\right).$$

We check this claim using the fact that $u+v \geqslant 2\sqrt{u}\sqrt{v}$ and $(u^p+v^p) \geqslant 2^{-(p-1)_+}(u+v)^p$ for any $p>0$ and $u,v\geqslant 0$. In this context, we have

$$\mathbb{W}_{\varrho_{exp}}(\mu_1,\mu_2) \leqslant \|\!|\mu_1-\mu_2|\!\|_{\varphi}.$$

The choice of the semi-metric ϱ in (1.10) is not unique. Indeed, developing the exponential function for any $q\geqslant 1$ we also have the estimate

$$\varphi(x_1)+\varphi(x_2) \geqslant \varrho_{p,q}(x_1,x_2) := c_{p,q}\ \|x_1-x_2\|^{pq} \quad\text{with}\quad c_{p,q} := 2c\ \frac{1}{q!}\left(\frac{b}{2^{1+(p-1)_+}}\right)^q, \tag{1.11}$$

and

$$c_{p,q}\ \mathbb{W}_{pq}(\mu_1,\mu_2)^{pq} \leqslant \mathbb{W}_{\varrho_{p,q}}(\mu_1,\mu_2) \leqslant \|\!|\mu_1-\mu_2|\!\|_{\varphi}.$$